MATH70013 Advanced Simulation Methods Coursework 2

This assignment will be done individually. Every student needs to deliver via blackboard **one** report with the answers to the questions. By submitting the coursework you automatically agree that this assessed coursework is your own work unless otherwise acknowledged, includes no plagiarism, you have not discussed it with anyone else except when seeking clarifications with the module lecturer, and have not shared any code with anyone else prior to submission. In your answers you can quote material from the notes and slides without need for replication or elaboration. The report should be in portable document format (PDF). Your report has to be uploaded to blackboard (URL: bb.imperial.ac.uk) at the *Coursework 2* container. The **deadline** is **Wednesday May 13th at 1pm**. You may use any programming language of your choice. The report will be marked from 0 to 100.

1 Filtering and smoothing for a nonlinear state space model

Consider the following state space model

$$X_n = \frac{1}{2}X_{n-1} + \frac{25X_{n-1}}{1+X_{n-1}^2} + 8\cos(1.2n) + V_n, \quad V_n \sim N(0, 0.1),$$
$$Y_n = \frac{X_n^2}{20} + W_n, \quad W_n \sim N(0, 1),$$

where $X_0 \sim \mathcal{N}(0, 1)$. Run the model to synthesise a data-set with $y_{0:T}$ for $T = 200$. Store the real state trajectory, $x^*_{0:T}$. Implement:

- (a) a single forward run of a SIR particle filter,
- (b) a dedicated particle smoothing algorithm.

You can choose which importance density $q(x_n|y_n, x_{n-1})$ will be used and for (b) your preferred smoothing method from fixed lag smoothing, FFBSm or FFBSa.

1. [11 marks] For (a) state your choice of $q(x_n|y_n, x_{n-1})$ and justify the reasons behind your chosen importance density. Mention what smoothing method you implement for (b) and its tuning parameters (i.e size of lag for fixed lag smoothing or number of sampled paths for FFBSa).
   For $n = 1, 50, 100, 150, 199$ only, plot the particle approximations for $p(x_n|y_{0:n})$, $p(x_n|y_{0:T})$ using (a) and (b) for $N = 50$ and 500. You may use either histograms or smooth kernel density estimates to estimate the filter's PDF. Comment briefly on the accuracy of the results. Is the terminology "smoother" for $p(x_n|y_{0:T})$ justified here?

2. [4 marks] For each (a) and (b) using $N = 50$ produce separate plots illustrating whether path degeneracy is present.

3. [4 marks] For $N = 50$ and 500, produce two plots, each of which displays for $n = 0, \ldots, T$:

   - x^*_n, the two particle approximations for $E[X_n|y_{0:T}]$ using (a) and (b) against n, and the particle approximation for $E[X_n|y_{0:n}]$; all against n.
   - (and in a separate figure) the estimated variances of $p(x_n|y_{0:T})$ for (a), (b) above as well as the one from $p(x_n|y_{0:n})$; all against n.

4. [5 marks] Comment on the results above highlighting

   - estimation performance for (a) and (b),
   - differences between filtering and smoothing in general,

- specific setup related to N, T and algorithm specifications,
- computational cost,
- model specific features.

5. [6 marks] For the stated model above, we wish to check numerically theoretical properties for the following smoothed additive functional:

$$\mathcal{S}_n^\theta = \int \left[\sum_{k=0}^{n} x_k \right] p_\theta \left(x_{0:n} | \, y_{0:n} \right) dx_{0:n},$$

Based on the course material, for each (a) and (b) state a theoretical property for an appropriate estimator for $\mathcal{S}_n^\theta$ as a function of n, N and other constants. Suppose we are interested to test these theoretical properties numerically and verify only the dependence on n. Using a total budget of at most 50 independent runs of each algorithm, suggest a procedure to numerically verify these properties. In your answer mention

- how the algorithms in (a) and (b) will be used and the relevant outputs,
- what quantities will be plotted or tabulated,
- how will one assess whether the theoretical property is validated.

6. [6 marks] Implement the above procedure, display your results and comment whether these numerical results can be used to justify the state theoretical property and why.

2 Likelihood inference using synthetic data

For the following scalar models

$$\text{Model 1} \quad X_n = \rho X_{n-1} + \tau V_n, \quad Y_n = X_n + \sigma W_n, \tag{1}$$

and

$$\text{Model 2} \quad X_n = \rho X_{n-1} + \tau V_n, \quad Y_n = X_n + \sigma \exp(\frac{X_n}{2}) W_n, \tag{2}$$

where in both cases $W_n, V_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, $X_0 \sim \mathcal{N}(0,1)$. For each model, synthesise a data-set with $y_{0:T}^1$ and $y_{0:T}^2$ respectively for $T = 100$, $\rho = 0.8, \sigma = 1, \tau = 0.5$. Let $\theta = \left(\rho, \sigma^2, \tau^2\right)$. Use the same real state trajectory $x_{0:T}^*$ in each case and store it for future comparisons.

For either $j = 1, 2$ we will use the following notations:

- $p_{\theta,1}(y_{0:T}^j)$ denotes the ideal marginal likelihood under Model 1.
- $p_{\theta,2}(y_{0:T}^j)$ denotes the ideal marginal likelihood under Model 2.
- $p_{\theta,1}^N(y_{0:T}^j)$ denotes the particle approximation $p_{\theta,1}(y_{0:T}^j)$ using standard SIR with a bootstrap proposal and N particles.
- $p_{\theta,2}^N(y_{0:T}^j)$ denotes the particle approximation $p_{\theta,2}(y_{0:T}^j)$ using standard SIR with a bootstrap proposal and N particles.

We will perform an interesting exercise whereby we will fit both datasets to both models for the sake of comparison. In both cases one will be the model that generated the data. Note here each $p_{\theta,j'}^N(y_{0:T}^j)$ should be thought as a function of θ to be maximised.

1. [9 marks] Assume just ρ, τ are known and set to their true values and only σ^2 is an unknown parameter and suppose $\sigma^2 \in [0.01, 2]$. Using a grid of values for σ^2 that uses 200 points and $N = 500$, produce the following plots
   (a) $\log p_{\sigma^2,1}(y_{0:T}^1)$ versus σ^2, $\log p_{\sigma^2,1}^N(y_{0:T}^1)$ versus σ^2, $\log p_{\sigma^2,2}^N(y_{0:T}^1)$ versus σ^2
   (b) $\log p_{\sigma^2,1}(y_{0:T}^2)$ versus σ^2, $\log p_{\sigma^2,1}^N(y_{0:T}^2)$ versus σ^2, $\log p_{\sigma^2,2}^N(y_{0:T}^2)$ versus σ^2
   (c) Suggest an estimate for the Maximum Likelihood Estimator (MLE) for σ^2 in each case.
2. [10 marks] Discuss the results using comments based on:

- intuition from the model dynamics, information from $y^1_{0:T}, y^2_{0:T}, x^*_{0:T}$,
- general understanding of the methods used,
- the quality of the resulting particle approximations and how they affect the results,
- simulations from the fitted models.

3. [3 marks] Suggest possible ways to refine the implementation of the method used above.

4. [10 marks] For Model 2 suggest an alternative iterative estimation method to compute the MLE that can be used either with standard SIR or a particle smoothing method. Implement your chosen method using $y^2_{0:T}$ with both SIR and particle smoothing and plot 100 iterations of σ^2 in each case using $N = 100$. Comment on the reason for possible differences in final estimate between the two and the grid method above.

3 Bayesian inference for a real data-set

In the file `data.csv` you can find daily log returns from recent exchange rate data between UK sterling and US dollar. (The time series contains $\log \frac{r_t}{r_{t-1}}$ with r_t being the closing spot exchange rate of day t). We wish to fit the following stochastic volatility model to this data:

$$X_n = \rho X_{n-1} + \sigma V_n, \quad Y_n = \beta \exp(\frac{X_n}{2}) W_n, \tag{3}$$

where $W_n, V_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, $X_0 \sim \mathcal{N}(0, 0.01)$. For the static parameters you may use any (sensible) prior of your choice, but a possible choice or starting point could be $\sigma^2 \sim IG(1,1)$, $\beta^2 \sim IG(1,1)$,and $\rho \sim U[-1,1]$.

1. [2 marks] Present a MCMC algorithm appropriate for this task. State what priors will be used. (You are permitted to use convenient initialisations and priors to accelarate MCMC convergence).

2. [10 marks] Implement this approach for the first 1000 data-points (i.e. we set $T = 1000$ to approximate $p(\rho, \tau^2, \beta^2 | y_{1:T})$). Present marginal densities for the static parameters of the model together with two diagnostics of your choice for each variable. State values for any tuning variables.

3. [10 marks] One is interested to perform prediction using these results. Mention how you can approximate the predictive posterior $p(y_{T+1:T'}|y_{1:T})$ where T' is the total length of the data-set. Make sure you include in your answer how this approximation is derived and any steps needed for implementing your approach.

4. [5 marks] Plot the 99% credible intervals for $p(y_{T+1:n}|y_{1:T})$ versus n for $T < n \leq T'$ together with the true data $y_{T+1:T'}$ and the posterior mean for $p(y_{T+1:n}|y_{1:T})$. For the credible intervals one may use cloud plots, plots of the $0.025, 0.975$-quantiles varying with time or any other similar sensible choice.

5. [5 marks] Based on all the above discuss whether this is an appropriate model for this data-set and why.